



Microeconomics AP Review Practice

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Range: AP Microeconomics range + Case study

Note: (Review Guide/Review Practices/Practice Set) are UNOFFICIAL sources of reference and do NOT include or hint at questions in the actual exam!

Multiple choice:

1. Bruce is a talented writer and graphic artist who enjoys both types of work equally. Instead of earning \$45,000 as a writer, Bruce now earns \$25,000 in accounting profits as a graphic artist using the same computer he would have used as a writer. What is Bruce's economic profit from choosing to work as a graphic artist?

- (A) -\$45,000
- (B) -\$20,000
- (C) \$20,000
- (D) \$45,000
- (E) \$70,000

2. The socially optimal quantity of a public good is provided when

- A) marginal private cost equals marginal private benefit
- (B) marginal social cost equals marginal social benefit
- (C) total private cost equals total social benefit
- (D) total social cost equals total social benefit
- (E) marginal social benefit minus marginal social cost is at a maximum

3. Game theory is most useful in describing outcomes in markets in which

- (A) firms are price takers
- (B) there is only one producer
- (C) there are many small producers
- (D) products are identical for all firms
- (E) there are interdependent firms

4. Which of the following is a characteristic of firms in a perfectly competitive industry?

- (A) Engaging in collusive behavior
- (B) Facing significant barriers to entry
- (C) Setting the market price
- (D) Producing identical products
- (E) Earning long-run economic profit

5. Command economies differ from capitalist economies mainly because of which of the following?

- (A) Command economies have higher prices for oil.
- (B) Capitalist economies have higher prices for oil.
- (C) Command economies have higher degree of government ownership of resources.
- (D) Capitalist economies have higher degree of government ownership of resources.
- (E) Command economies rely on prices to allocate resources.

6. Consider the table below. If the average variable cost is \$60 for an output of 11 units, then

<u>Total Output</u>	<u>Total Cost</u>
0	\$100
2	196
4	212
6	310
8	430
10	570

- a. the total variable cost is \$630.
- b. the total cost of 11 units of output will be \$652.
- c. the total cost of 11 units of output will be \$660
- d. the marginal cost of the eleventh unit is \$102.
- e. the marginal cost of the eleventh unit is \$190.

7. In the long run, firms in monopolistic competition typically earn

- A) Economic profits due to barriers to entry.
- B) Zero economic profits due to free entry and competition.
- C) Losses because of excessive advertising costs.
- D) Monopoly-level supernormal profits.
- E) infinite profit

8. An example of a negative externality is the impact of

- a. bad weather on the incomes of farmers.
- b. purchasing Netflix Premium.
- c. pollution from a factory on the health of people living nearby the factory.
- d. increases in health care costs on the health of individuals in society.
- e. taking vaccines.

9. In monopolistic competition, advertising is primarily used to:

- A) Lower production costs through economies of scale.
- B) Make demand curves perfectly inelastic.
- C) Create brand loyalty and perceived product differentiation.
- D) Comply with government regulations.
- E) Decrease profit

10. The kinked demand curve model of oligopoly suggests that firms:

- A) Ignore competitors' pricing decisions.
- B) Match price cuts but not price increases, leading to price rigidity.
- C) Always act as price takers.
- D) Produce at minimum average total cost in the long run.
- E) Are likely to become monopoly

11. In perfectly competitive markets, the market equilibrium wage results in the most efficient outcome. In turn, setting a minimum wage reduces the total social surplus. Labor unions that negotiate a wage above what the employer wants to offer are a cause of deadweight loss. However, when the markets are not perfectly competitive, higher wages may be beneficial for the economy. From Microeconomics perspectives (i.e., ignoring the effect on Aggregate Demand), for which of the following market imperfections higher wages may, most likely, lead to higher total social surplus?

- A) Monopoly
- B) Monopsony
- C) Monopolistic competition
- D) Bertrand Oligopoly (i.e., markets where oligopolists compete with each other by setting the price)
- E) Cournot Oligopoly (i.e., markets where oligopolists compete with each other by choosing

the production quantity)

12. Two straight-line demand curves intersect each other at the point of intersection. How does the price elasticity of demand compare between these two curves at the point of intersection?

- A) The price elasticity is higher on the flatter curve than on the steeper curve.
- B) The price elasticity is the same for both curves at the point of intersection.
- C) The price elasticity is higher on the steeper curve than on the flatter curve.
- D) The price elasticity is zero at the point of intersection for both curves.
- E) The price elasticity is infinite at the point of intersection for both curves.

13. Which of the following policy is used to control negative externalities?

- a. Subsidy
- b. Income Policy
- c. Tax
- d. Monetary policy
- e. Fiscal policy

14. An entrepreneur leaves a £50,000/year job to start a business. The lost salary is an example of:

- A) An explicit cost, because it was a direct monetary payment.
- B) An accounting cost, because it appears on financial statements.
- C) An implicit cost, because it represents an opportunity cost.
- D) A opportunity cost, because it represents the sum of all of alternative decisions the entrepreneur could have made.
- E) An investment cost, because it causes the entrepreneur to take the risk of not getting larger or equal amount of profit.

15. In a perfectly competitive market, firms are price takers because

- A) They collude to set prices artificially high.
- B) Barriers to entry prevent new firms from competing.
- C) Each firm is too small to influence the market price.

D) Products are highly differentiated.

16. A monopolist with a demand function $Q = 120 - P$ and total cost $TC = 2Q^2$ faces per-unit tax of \$12. By how much the consumer price will increase?

- A \$3
- B \$6
- C 2%
- D 10%

17. Let BMW and Mercedes be the only two competitors in the automobile industry. All else equal, if Mercedes reduces the price of its cars, we should expect that

- A BMW's supply curve will move to the left and the price of BMW cars will increase
- B BMW's supply curve will move to the right and the price of BMW cars will decrease
- C BMW's demand curve will move to the left and the price of BMW cars will decrease
- D BMW's demand curve will move to the right and the price of BMW cars will increase

18. Let's suppose that Catherine just lost her job and found a new one right away but at a lower wage rate. Based on the neoclassical theory for labor supply, the substitution effect of the lower wage for Catherine will lead her to:

- A work less hours than before
- B work more hours than before
- C work the same hours as before
- D exit the labor market and thus, supply zero hours of work

19. If a producer has market power (that is, with her actions she can influence the market price),

what can we say about the quantity produced?

- A It is equitable
- B It is efficient
- C It maximizes consumer surplus
- D None of the other answers is correct

20. Consider the Rock-Paper-Scissors game with the following payoffs. Which of the following is true?

			
	0 : 0	-1 : 1	1 : -1
	1 : -1	0 : 0	-1 : 1
	-1 : 1	1 : -1	0 : 0

- A Losing this game is just as hard as winning
- B The game has one pure-strategy Nash equilibrium
- C The game has three pure-strategy Nash equilibria, each with (0 : 0) payoff
- D The game has no pure- or mixed-strategy Nash equilibria

21. Suppose that Jamal purchases two goods X and Y. The marginal utility of X is MU_x , the total utility of X is TU_x , the marginal utility of Y is MU_y , and the total utility of Y is TU_y . If the prices of X and Y are P_x and P_y respectively, which of the following expressions defines consumer equilibrium?

- (A) $TU_x = TU_y$
- (B) $MU_x = MU_y$
- (C) $TU_x/P_x = TU_y/P_y$
- (D) $MU_x/P_x = MU_y/P_y$
- (E) $(MU_x)(P_x) + (MU_y)(P_y) = 1$

22. Assume that total fixed costs are \$46, that the average product of labor is 5 units when 10 units of output are produced, and that the wage rate is \$12. If labor is the only variable input, what is the average total cost of producing 10 units of output?

- (A) \$ 2
- (B) \$ 5
- (C) \$ 7
- (D) \$ 9
- (E) \$12

23. Which of the following best explains why it is difficult to maintain lasting collusive agreements?

- (A) There is an unavoidable conflict in that a collusive agreement can increase the profits of

some, but not all, firms in the industry.

(B) There is little potential for gain from collusion unless there is a large number of consumers in the market.

(C) Each firm in the industry views itself as facing a vertical demand curve, even though the market demand curve is downward sloping.

(D) The firms in the industry have a common incentive to increase output to a more competitive level.

(E) Each firm realizes that its profits would increase if it were the only firm to violate the collusive agreement by increasing its production slightly.

24. If labor is the only variable input and it costs \$15 per hour and if the marginal product of labor is 3 units per hour, the short-run marginal cost of 1 unit of output is approximately

(A) \$ 0.20

(B) \$ 3.00

(C) \$ 5.00

(D) \$15.00

(E) \$45.00

25. If a perfectly competitive firm wishes to maximize profits and is producing where price exceeds both marginal cost and average variable cost, then the firm is

(A) producing too little output

(B) producing where its marginal revenue is less than its marginal cost

(C) using too much of the variable input

(D) charging too high a price for its output

(E) not earning any economic profits

26. The condition for allocative efficiency is violated when

(A) firms are price makers (price searchers)

(B) short-run profits exist in a competitive industry

(C) price equals average total cost

(D) the market demand curve is inelastic in a competitive industry

(E) the market demand curve is elastic in a competitive industry

27. In which of the following cases is the government's action appropriate for reducing

inefficiency?

- (A) Taxation of the output of a chocolate factory that emits an aroma that residents of the city enjoy
- (B) Regulation that reduces the output of a monopoly
- (C) A lump-sum tax on a monopolist
- (D) A subsidy for the consumption of a good produced by a plant that emits a pollutant as a by-product
- (E) A toll on a congested bridge

28. The concept of derived demand is described by which of the following?

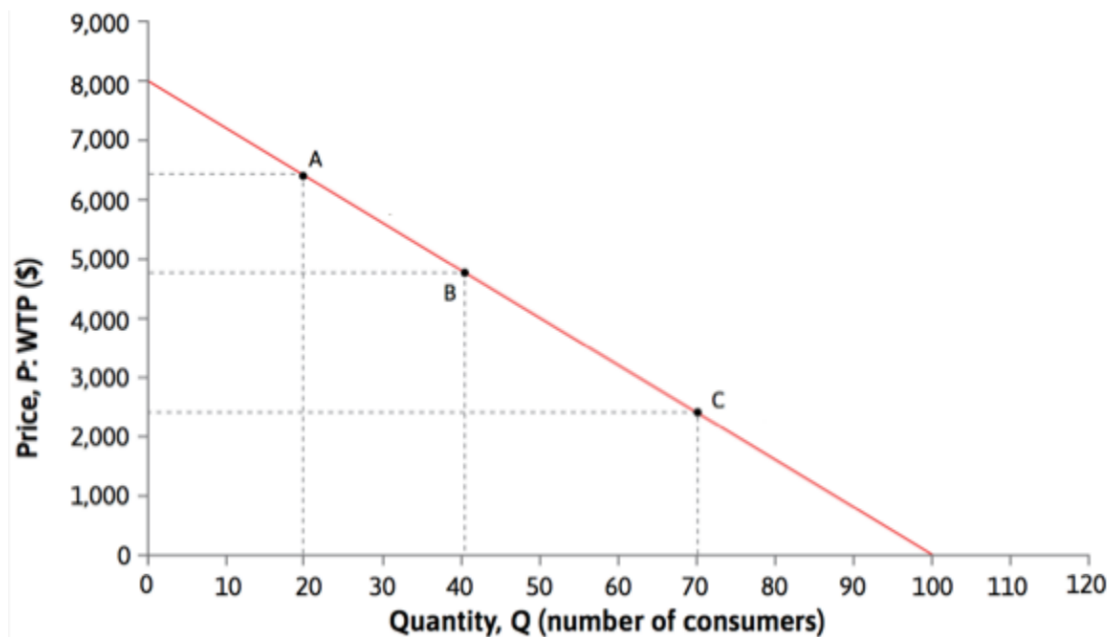
- (A) A decrease in the demand for theater tickets will decrease the demand for actresses and actors.
- (B) If the salaries of basketball players increase, the quantity of basketball players demanded will decrease.
- (C) An increase in the income of consumers will increase the demand for opera tickets.
- (D) An increase in the demand for movie tickets will decrease the demand for video rentals.
- (E) A decrease in the price of movie tickets will increase the demand for movie tickets.

29. Which of the following is the best example of a negative externality?

- (A) An increase in the price of oil due to the imposition of environmental regulations
- (B) An increase in the price of oil due to action taken by the Organization of Petroleum Exporting Countries (OPEC)
- (C) A decline in oil stock prices as a result of bad management
- (D) Oil leakages from drilling platforms in the Gulf of Mexico
- (E) Declining restrictions on the importation of foreign-made car

30.

The demand curve of a price-setting firm is shown below.



Price-setting firm (monopoly) may find it optimal to produce at

- A. points B or C only
- B. points A or B only
- C. points A, B, and C
- D. point A only

Free Response Question:

1. Sugar is produced in a perfectly competitive market using inputs from perfectly competitive factor markets. Frank Sugar Co. is a representative firm in the sugar market.

(a) Assume Frank Sugar Co. is earning zero economic profit. Draw correctly labeled side-by-side graphs for the sugar market and Frank Sugar Co. and show each of the following.

(i) The market equilibrium price and quantity, labeled P_M and Q_M , respectively

(ii) The profit-maximizing price and quantity for Frank Sugar Co., labeled P_P and Q_F , respectively

(b) Assume the demand for sugar increases and sugar is produced in a constant-cost industry.

(i) On your graph in part (a), show the short-run effect of the increased demand for sugar on the market price, labeled P_2 , and the quantity sold by Frank Sugar Co., labeled Q_N

(ii) Compared to the equilibrium identified in part (a)(ii), what will happen to the short-run profit earned by Frank Sugar Co. as a result of the increased demand for sugar?

(iii) When the market adjusts to long-run equilibrium, how will the market price of sugar in the

long run compare to P2? Explain.

(c) Instead, assume sugar consumption has a negative impact on public health over time and the negative impact of sugar on health is underestimated by consumers. Draw a correctly labeled graph of the market, with the marginal social benefit (MSB), marginal private benefit (MPB), marginal social cost (MSC), and marginal private cost (MPC) curves, and show each of the following.

(i) The market equilibrium quantity, labeled QM

(ii) The socially optimal quantity, labeled Qs

(d) Assume the government decides to intervene in the market to affect consumers' incentives and to address the negative impact of sugar consumption on public health. Which of the following policies would best achieve that objective in the short run: a lump-sum tax, a per-unit tax, a lump-sum subsidy, or a per-unit subsidy? Explain.

Case Study

Is It Worth Breaking Out the World Trade War?

Author: Teresa Kamińska

The trade liberalization tenet has governed international economic relations since the Second World War, which has contributed the urging of integration and globalization processes. The United States of America used to play the role of the main leader in international organizations (mostly The United Nations, World Bank and World Trade Organizations) that created order in global trade. As a result of never-ending negotiations targeting the gradual lowering of duties and other fiscal and non-fiscal tariffs, as well as various barriers, it was roughly possible to solve or at least mitigate disputes between countries. The sudden rebuttal of the so far trade policy in the late 2010s by the USA seemed to “turn over the boat”. A new doctrine relying on the adage, “America first,” means paralyzing everything in the frames of the WTO, which might be disadvantageous for the American economy or its world position in general. In order to keep a powerful standing while negotiating new rules, the American government refuses to talk with groups of countries (e.g. The European Union, NAFTA) and insists on debates bilaterally. The battle began with imposing a significant import tariff for steel (25%) and aluminum (10%) if exporting countries did not voluntarily reduce their exportation of those commodities to the USA. Such a political game succeeded up-to-now (August 2018) as related to three countries (South Korea, Brazil and particularly Mexico, although in recent years some steelworks have been moved there). However, nowadays there are other meaningful American trade partners, i.e. Canada, the European Union and China, which are not so submissive. They differ from each other regarding economic targets, which results in a lack of leadership, but to be effective they should create an alliance. Meanwhile the EU, China and Canada prepared a list of retaliation charges. The effect was surprising because American producers (e.g. Harley – Davidson) threatened to move some of their products to its factories

elsewhere abroad. In turn, the American government threatened companies with imposing extra taxes for such a disadvantageous activity.

Considering the EU, President Trump's policy is mainly aimed at the German automobile industry/sector. Substantial increasing of import tariffs for automobiles (BMW, Volkswagen and Daimler) will result in their price soaring in the American market, although they already have their own car factories there. Eventually, Germany and the network of car component producers in other EU countries will lose. If this happened, the EU would implement severe retaliations.¹⁸ American – Chinese clashes pertain to an immense trade surplus that has existed for decades since investors – not only Americans – transformed China into the “world factory”. However, the ¹⁸ It is being estimated that Germany might lose 0.2% of GDP, and Poland 0.16% of GDP. The UE's retaliations might induce an unemployment increase in the USA as much as about 4 million employees. The problem is more combined because many accusations displayed by the American government against Chinese activities in various fields aimed at reducing the increasing world power of the Chinese economy (i.e. joint ventures, illegal copying of new technologies and products, investments in IT sector). Finally, the USA has gradually been raising tariffs for Chinese merchandise and the latter country replied in the same way (the list of Chinese retaliations seems to be longer including financial instruments and political actions weakening sanctions against the North Korea). The problem remains since Chinese imports are still dissatisfactory.

Summing up, the American policy is not transparent because of its discrepancies in performance (e.g. breaking negotiations of creation Trans-Atlantic Trade and Investment Partnership – TTIP). Trade partners like the EU, Japan, or Latin America do not want to follow the US performance, principally as they expect advantages from American – Chinese trade conflicts.

Questions

1. Discuss the possible short and long-run results of the introduction of protectionism to international trade.
2. Graphically illustrate the impact of tariff increases on imports of steel and aluminum to the US on:
 - a) The US submarket of imported steel and aluminum, heavy industry and labor market
 - b) markets (exportation sectors) in the following countries and their labor markets:
 - China
 - The EU
3. If the German automobile imports were burdened by much higher duties than at present (25–30% instead of 2.5%) what countries would suffer?